

Disinformation, AI, and Regulation in Ecuador's 2025 Presidential Election

1. Core Objective & Scope

The study analyzes Ecuadorian citizens' and legal/communication experts' perceptions regarding the deployment of **generative artificial intelligence (AI)** to spread disinformation during the 2025 presidential elections. It evaluates the subsequent structural risks to democratic integrity, institutional trust, and the looming necessity for structured regulatory frameworks.

2. Methodology & Demographics

The paper utilizes a mixed explanatory approach combining quantitative and qualitative methodologies:

- **Quantitative Data:** Two non-probabilistic surveys distributed via a "snowball" dynamic on Google Forms.
 - *Survey 1 (Pre-election, Nov–Dec 2024):* n = 201 participants (predominantly aged 18–27 from the highland region).
 - *Survey 2 (Post-election, Jan–Feb 2025):* n = 64 participants, collecting open-ended qualitative data.
- **Qualitative Data:** 14 semi-structured interviews via Zoom with experts in law, academic research, and electoral administration (conducted both before and after the elections).
- **Sentiment Analysis:** Manual analysis of the 64 open-ended survey responses to map emotional demographic patterns.
- **Technical Note:** The author utilized **Qwen3-Max** to organize interview transcripts and structure the sentiment analysis.

3. Key Statistical Findings (Pearson’s Chi-Square Tests)

The quantitative analysis tested three main hypotheses evaluating variables against the use of AI in electoral campaigns:

Hypothesis Tested	Chi-Square Metric	Significance (p-value)	Outcome / Strength

H1: Association between <i>compliance with AI regulation and perception of the election outcome.</i>	chi ² = 9.298, gl=3	p = 0.26	No statistically significant association found.
H2: Association between <i>AI campaign usage and the level of trust in candidates' ads.</i>	chi ² = 27.095, gl=6	p < 0.001	Statistically significant association (Modest strength: Cramer's V = 0.26).
H3: Association between <i>AI campaign usage and trust in the final election results.</i>	chi ² = 18.833, gl=6	p = 0.04	Statistically significant association (Modest strength: Cramer's V = 0.216).

Takeaway: The empirical data demonstrates that the usage of AI in campaigns directly impacts and depresses public trust in both candidate advertisements and final voting outcomes.

4. Empirical Evidence of AI Abuse

Data provided by the local organization *Usuarios Digitales* mapped 36 verified tactical cases of AI manipulation during the official campaign window (Jan 18 - Apr 2, 2025):

- **Tactics Used:** Synthetic images, inauthentic videos with voice cloning, and synthetic videos featuring deepfakes.
- **Intent Distribution:** 9 posts were used for *Support*, 18 for *Attack* (black campaigns targeting frontrunners Luisa González and Daniel Noboa), and 9 for *Satire*.
- **Primary Vectors:** TikTok (21 cases) and X (formerly Twitter; 9 cases) were the primary platforms utilized for spreading synthetic media.

5. Regulatory Gaps & Legal Analysis

From a legal framework perspective, the paper highlights a critical mismatch between rapid technological evolution and Ecuadorian jurisprudence:

- **The Legislative Loophole:** While several draft laws to regulate AI had been introduced to the National Assembly of Ecuador by 2025, **none of them explicitly addressed electoral risks**. They instead focused broadly on data privacy, facial recognition, autonomous vehicles, and healthcare systems.
- **Institutional Impotence:** Experts noted a profound "legal loophole" in Ecuador's *Democracy Code*. The National Electoral Council (CNE) completely lacked the technical infrastructure and legal mechanisms required to issue rapid takedown requests to foreign social media platforms or effectively sanction actors using synthetic media. Consequently, regulatory effectiveness during the 2025 cycle was practically nil.

6. Sentiment & Public Demands

Manual sentiment tracking revealed that **60% of citizens experience concern/fear** and **55% express deep distrust** toward AI in politics, viewing it as a structural threat to democratic consensus.

Despite this, the public is highly proactive, overwhelmingly demanding a **co-regulation or regulated self-regulation model**. Rather than an outright ban on AI, public and expert consensus points to a framework led by the CNE incorporating:

1. **Mandatory Labelling:** Legally requiring watermarks or disclosure messages specifying content is AI-generated.
2. **Legal Penalties:** Establishing explicit economic or political sanctions for malicious disinformation.
3. **State-Sponsored Digital Literacy:** Integrating a structural "culture of verification" directly into the public education system to build cognitive resistance to deepfakes.